

MatKeyframer

File path: Assets/UIAnimationToolkit/Runtime/Materials/MatKeyframer.cs

Assets/UIAnimationToolkit/Editor/Materials/MatKeyframerEditor.cs

Menu: Add Component > UI Animation Toolkit > Mat Keyframer

Overview

Expose selected UGUI shader properties as animatable component fields. The runtime component implements `IMaterialModifier` to create a dedicated material instance and applies slot values during editing and playback. The custom Inspector reads supported visible shader properties and lets you enable them without manually filling every slot name.

How to use

1. Select an Image, RawImage or other object with a UGUI Graphic. Assign a material with the shader properties you want to animate.
2. Add Mat Keyframer. Its Inspector currently displays the heading Mat Anim Wrapper.
3. Expand Inactive and enable a property. The Inspector assigns a compatible free slot and initializes its value from the material.
4. Edit enabled properties under Active. Repeat for other properties, staying within the slot capacities below.
5. In the Animation window, add/record the corresponding MatKeyframer value fields on the object, then create keys. Slot field names are listed below for identification.
6. Preview the animation. Disable a property only when you no longer want that slot assignment; verify existing animation bindings if you reassign slots.

Interface and parameters

Mat Anim Wrapper

Active	Enabled shader properties
Float example	_Intensity -> f1Value
Color example	_Tint -> c1Value
Texture	Off = offset Sc = scale
Inactive	Expand to enable more properties

Inspector schematic. Shader property names shown here are illustrative, not bundled shader features.

Control / parameter	Description
Float / Range	4 slots: f1Name-f4Name with f1Value-f4Value.
Color	4 slots: c1Name-c4Name with c1Value-c4Value.
Texture transform	2 slots: t1Name/t2Name with t1Offset/t2Offset and t1Scale/t2Scale.
Vector4	2 slots: v1Name/v2Name with v1Value/v2Value.
Active	Properties currently assigned to slots. Edit their values here.
Inactive	Supported properties not assigned; check a row to allocate a slot.
Property checkbox	Enable/disable assignment. If all matching slots are used, the Inspector reports the limit.
No Material warning	Assign a Graphic material; the Inspector falls back to manual slot fields when no source material is available.

Notes

- This component is for UGUI Graphic objects, not a general MeshRenderer/SkinnedMeshRenderer material animation system.
- Texture slots animate offset and scale; they do not animate texture asset replacement.
- Visible shader names/descriptions come from the assigned shader. The toolkit does not translate user-authored shader labels.
- Each component manages a material instance, which can affect batching and material count. Test Mask/Stencil combinations, material switching and component enable/disable in the target project.
- The runtime class is UIAnimationToolkit.MatKeyframer. Keep its script in Runtime and its custom Inspector in Editor.
- Inspector edits record Undo. No universal Unity-version, shader or render-pipeline compatibility is implied by this documentation.